

Description of files in this folder

1. run_trim_galore.sh
 - a. Command used to run TrimGalore on the samples
 - b. FASTQs are available under NCBI BioProject ID PRJNA1328687
2. run_kallisto.sh
 - a. Command used to run kallisto after trimming
3. sample_info_transcriptomic_screen.xlsx
 - a. Used as metadata for the scripts below
4. pca.Rmd
 - a. Code used to produce PCA plots for the in vitro transcriptomic screen
 - b. Inputs:
 - i. The kallisto files which are available on GEO
 - ii. sample_info_transcriptomic_screen.xlsx (in this folder, see above)
 - c. Outputs – six PDFs in the pca_plots subfolder (used for multiple supplementary figures)
5. deseq2.Rmd
 - a. Code used to run DESeq2 for the in vitro transcriptomic screen
 - i. Inputs:
 1. The kallisto files which are available on GEO
 2. Reference transcriptome downloaded from Ensembl (release 110, Homo_sapiens.GRCh38.cdna.all.fa.gz)
 3. sample_info_transcriptomic_screen.xlsx (in this folder, see above)
 - ii. Outputs:
 1. DESeq2 outputs in the deseq2_outputs subfolder (used for supplementary tables)
 2. A supplementary figure showing the # DEGs per metabolite
 3. Correlation plots showing that vehicle controls that had the same solvent in the media, but were on different plates within a batch, could be grouped
6. lps_up_kegg_pathways_affected.Rmd
 - a. Inputs:
 - i. gprofiler outputs batch1-batch4.csv which are contained in the lps_up_gprofiler_kegg subfolder
 1. These are gprofiler outputs based on the genes that LPS upregulated (see deseq2_outputs subfolder, above)
 2. These were used to create one of the supplementary tables
 - b. Outputs:

- i. PDFs batch_1 through batch_4 in that same subfolder (used for a main figure panel and a supplementary figure)
- 7. num_degs_in_ad_pd_kegg_pathways.Rmd
 - a. Inputs:
 - i. DESeq2 outputs (please see above)
 - ii. Lists of genes in Alzheimer's Disease / Parkinson's Disease KEGG pathways, downloaded from <https://rest.kegg.jp/link/hsa/path:hsa05010> and <https://rest.kegg.jp/link/hsa/path:hsa05012> on July 19, 2025
 - b. Outputs:
 - i. A list of DEGs that overlap the AD/PD KEGG pathways, for each metabolite, within the ad_pd_genes_affected subfolder
 - ii. A figure panel showing the # AD/PD genes affected per metabolite
- 8. chem_structure_svgs.py
 - a. No inputs; outputs the four structure diagrams shown in main Fig. 2
 - i. These outputs are present in this folder as:
 - 1. tmao.svg
 - 2. iaa.svg
 - 3. palmitate.svg
 - 4. gdca.svg
 - b. rdkit must be installed to run this script